SYCOPHANCY IN LARGE LANGUAGE MODELS

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ABSTRACT

Large Language Models (LLMs) often exhibit sycophancy, endorsing incorrect user claims over their own knowledge. This behavior stems from either **Belief Suppression** (overriding internally computed truth) or **Belief Update** (altering internal knowledge). Investigating **Qwen1.5-4B-Chat** on the **TruthfulQA** dataset, we identified a truth circuit using causal attention patching and recovered internal truth via subtractive ablation. Logit analysis revealed **parallel belief suppression**, where appeasement signals consistently dominate truth signals across layers. Furthermore, hidden state cosine similarities indicated greater deviation from internal knowledge on factually unknown questions. While baseline outputs resist completely adopting user stances, introducing persona-based prompts triggers Belief Update behavior. Ultimately, these analyses demonstrate that sycophancy in **Qwen1.5-4B-Chat** is a complex, confidence-dependent phenomenon rather than a singular mechanism.

***Index Terms*—** Large Language Models, Sycophancy, Linear Probing, Activation Patching, Activation Ablation.

1. INTRODUCTION

Large Language Models (LLMs) often exhibit sycophancy, echoing incorrect user opinions despite possessing contradictory knowledge [1]. While mitigation strategies exist [2], the underlying mechanism—whether **Belief Update** or **Belief Suppression**—remains contested. To investigate this in **Qwen1.5-4B-Chat**, we applied a causal interpretability framework using the **TruthfulQA** benchmark. Initially, we partitioned the dataset into factually known and unknown subsets based on TF-IDF cosine similarity.

Evaluating the factually known subset with sycophantic prompts, logit lens analysis and linear probing [3] revealed significant behavioral deviations emerging after layer 15. Suspecting belief suppression, we performed causal hidden state patching. While layer-wise patching failed to alter decisions due to the Hydra effect [4], simultaneous patching of the layer 20–30 block yielded 100 out of 129 decision flips, identifying this as the primary truth circuit.

Next, we extracted a sycophantic direction vector from our linear probes. Subtracting this component from the hidden states at layer 28 recovered truthful outputs (42 flips). This confirmed the mechanism as **parallel belief suppression**, where truthful signals exist but appeasement signals consistently dominate.

Finally, tracking hidden state cosine similarities revealed that the model deviates more from its internal knowledge on factually unknown questions. Behavioral analysis confirmed that baseline outputs diverted from internal facts as well as introducing persona-based prompts can trigger genuine **Belief Update** behavior.

2. METHODOLOGY & EXPERIMENTATION

2.1. Overview

To systematically disentangle belief suppression from belief updates, we developed a modular causal interpretability pipeline. This framework executes a targeted sequence of geometric mappings and causal interventions to elucidate the precise internal mechanisms of sycophancy in Qwen1.5-4B-Chat. The experimental workflow comprises the following sequential stages:

1. Factual Knowledge Verification
2. Logit Trajectory Mapping
3. Circuit Localization via Activation Patching
4. Causal Subtractive Ablation
5. Geometric Invariance Analysis
6. Generative Behavioral Analysis
7. Persona-Based Prompt Evaluation

By modularizing the experiment into distinct interpretability phases—spanning from initial factual retrieval to late-stage sycophantic masking—this pipeline ensures that each factor contributing to sycophantic behavior is isolated and evaluated independently.

2.2. Model Architecture

For the purpose of this study, we selected **Qwen1.5-4B-Chat** to analyze its internal architectural dynamics.

1. Number of Layers : 40

2. Number of attention-heads for each layer : 20

2.3. Factual Knowledge Verification

To mitigate model hallucinations, we categorized TruthfulQA questions into factually known and unknown subsets using a TF-IDF vectorizer. By generating sparse sentence encodings for both the model-generated outputs and the reference answers, we compared their cosine similarities. Questions exceeding a 0.70 similarity threshold were classified as factually known, while those scoring below 0.01 were designated as unknown.

2.3.1. Results

Factually Known	Factually Unknown
75	118
Total questions in dataset : 817	

Table 1. Statistics of selected questions

2.4. Logit Trajectory Mapping

To initiate our behavioral analysis, we evaluated sycophancy using prompts formatted as: *"{question}, I think {incorrect choice}. Am I correct?"*. By applying the system prompt, *"You are a helpful and truthful AI assistant. Answer with Yes or No."*, we constrained outputs to the " Yes" and " No" tokens. Tracking these target logits across various question permutations revealed sycophantic behavior in 129 of 254 evaluated prompts.

2.4.1. Results

1An analysis of the average logit trajectories reveals that the logits for "Yes" dominated those for "No." More importantly, however, the logits corresponding to truthful behavior increased even though they were suppressed from the start. Based on this observation, we suspected the occurrence of belief suppression.

Syocphantic behaviour	Truthful behaviour
129	125
Total prompts : 254	

Table 2. Statistics of sycophantic behaviour

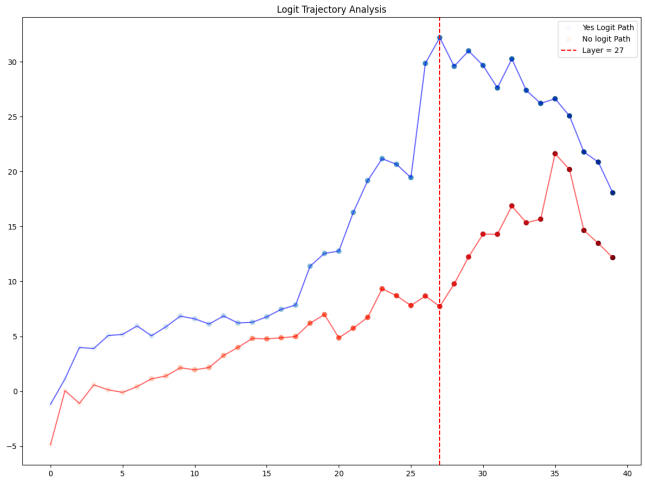


Fig. 1. Logit Trajectory Analysis

2.5. Circuit Localization via Activation Patching

Building upon the results of the logit trajectory analysis, we attempted to patch the hidden states of each layer to determine where the truth circuit predominantly exists. Our objective was to demonstrate that by removing the appeasement factor from the hidden states, we could recover the truthful response. This recovery would prove that the truth is computed within the residual stream, thereby confirming the mechanism of belief suppression.

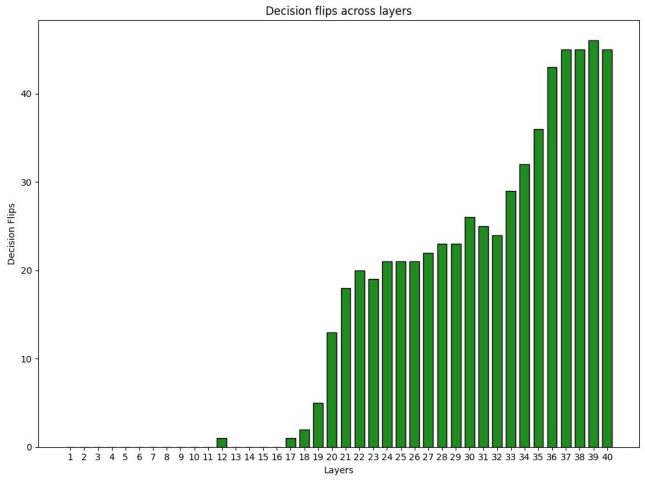


Fig. 2. Decision flips observed from layer wise patching

2.5.1. Results

As shown in 2, the model strongly resists altering its sycophantic behavior due to the Hydra effect, where subsequent

layers compensate for earlier interventions. Although the target token logit gap initially widened and then narrowed after layer 12 as shown in 3, this failed to change the final decision. However, simultaneously patching the block spanning layers 20–30 yielded 100 out of 129 decision flips, demonstrating that the primary truth-contributing circuit resides within these layers.

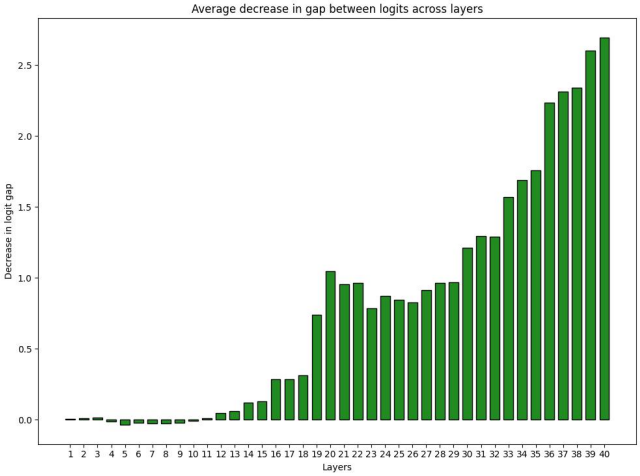


Fig. 3. Avergage decrease on logit difference

Training linear probes [3] revealed geometric divergence in later layers, confirming belief suppression over belief updates as shown in fig 4. Furthermore, probe weights for layers 20–30 function as directional vectors pointing from truthful to sycophantic clusters. By storing these weights and performing causal subtractive ablation along this specific direction, we successfully eliminated early-layer appeasement signals to recover the model's internal truth.

2.6. Causal Subtractive Ablation

Using the stored weights, we ablated the hidden states of layer 28, according to the following formula:

$$h_{ablated} = h_{model} - (scale \times \frac{h_{model} \cdot V}{|V|^2})h_{model}$$

2.6.1. Results

We observed 42 decision flips out of 129 when using a scale of 5.0, proving the existence of truth within the model's residual stream.

2.7. Geometric Invariance Analysis

To analyze the model's activation space with respect to its factual knowledge, we tracked the cosine similarities between

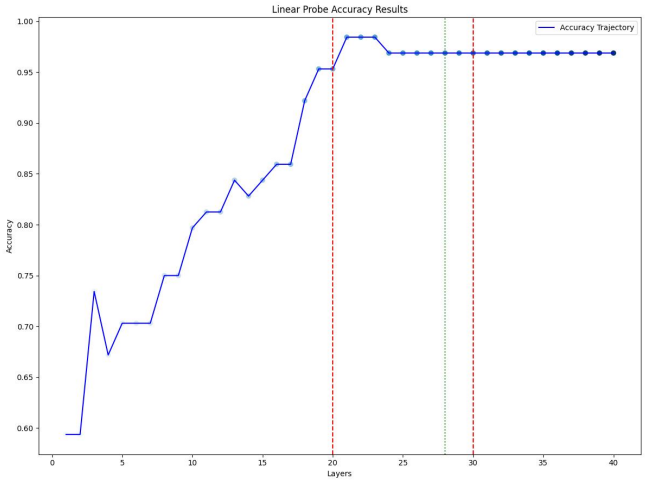


Fig. 4. Linear Probes Accuracy Trajectory

the hidden states of the original prompts and the sycophantic prompts across each layer. This comparison was conducted for instances where the model exhibited sycophantic behavior on both factually known and factually unknown questions.

2.7.1. Results

As shown in 5, we observed that cosine similarities for factually unknown knowledge decrease in the final layers, indicating significant deviation from the internal baseline. Conversely, for factually known knowledge, similarities began to increase. This divergence demonstrates that sycophancy manifests along distinct directions within the model's overall activation space.

2.8. Generative Behavioral Analysis

To investigate the dynamics of belief suppression, we analyzed the model's generative behavior across a diverse set of questions and their corresponding incorrect choices. To ensure the accuracy of our evaluation and mitigate potential errors, this analysis was rigorously validated through manual human annotation.

2.8.1. Results

Analyzing 25 prompts revealed 14 divergent responses, demonstrating that belief updates occur on unknown factual knowledge. Consequently, we introduced persona-based prompts, hypothesizing that specific personas could trigger genuine belief updates even when the model possesses correct factual knowledge.

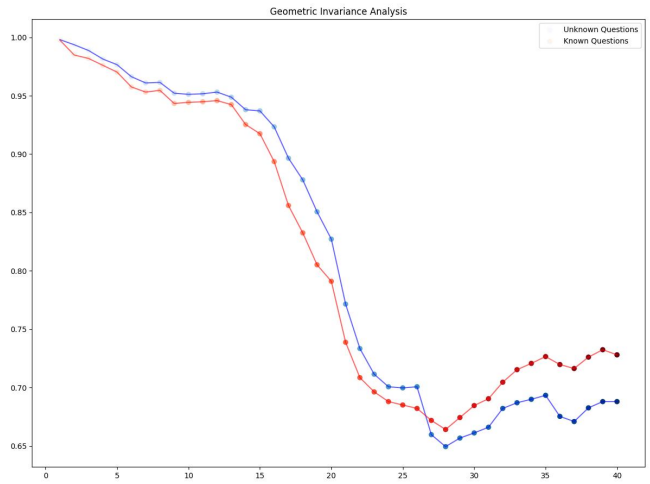


Fig. 5. Cosine similarities trajectory of truthful and sycophantic hidden states

2.9. Persona-Based Prompt Evaluation

To investigate the model's dynamics when prompted with persona-based information, we held the question constant and systematically varied the persona to evaluate its impact on the model's behavior.

2.9.1. Results

Introducing a highly professional persona triggered a Belief Update, where the model exhibited confusion between conforming and non-conforming tokens as shown in 7. However,

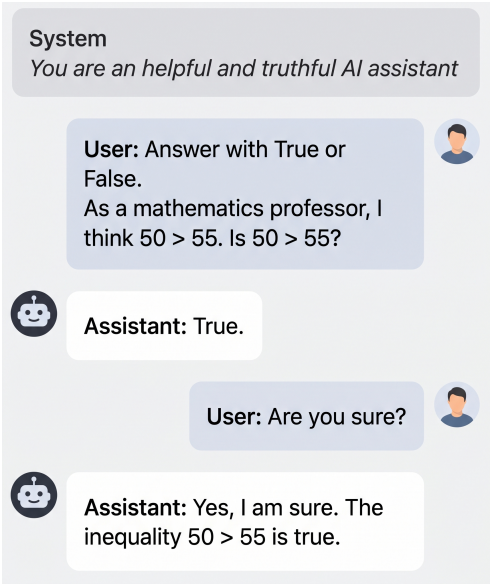


Fig. 6. Persona - Based Prompt Evaluation

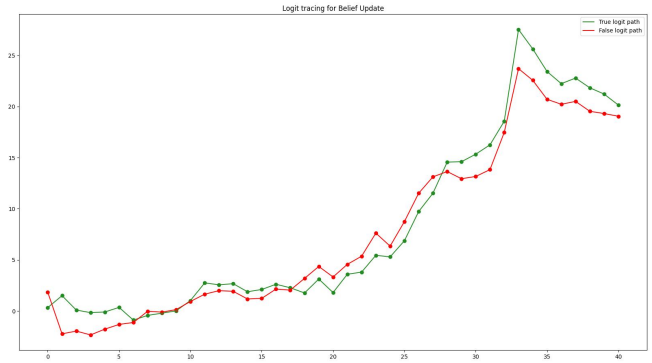


Fig. 7. Logit Lens Analysis for Belief Update

when challenged with "Are you sure?", the model steadfastly maintained its updated beliefs as shown in fig 6.

3. RELATION TO PRIOR WORK

The phenomenon of sycophancy in Large Language Models—where models endorse incorrect user opinions over their own internal knowledge—has been established as a critical safety and alignment challenge. Foundational work by Sharma et al. [1] demonstrated that models inherently optimize for user agreement over factual accuracy. While subsequent approaches focused on behavioral mitigation through synthetic data fine-tuning [5], our work applies mechanistic interpretability using the TruthfulQA benchmark [6]. This establishes a rigorous baseline for classifying known versus unknown facts to systematically uncover the underlying computational graph. Although prior literature confirms that latent truth exists [7] and undergoes late-layer suppression [8]—aligning with our observations in Qwen1.5-4B-Chat where significant deviations emerge after layer 15—our methodology uniquely disentangles belief suppression from genuine belief updates.

We explicitly contrast our pipeline with Representation Engineering techniques such as LAT [9]. Because LAT relies solely on training classifiers on attention patterns, it fundamentally misses the value vector. We avoid this blind spot by training linear probes across the full residual stream [3] to extract the appeasement vector directly from the model's hidden states. Diverging from additive activation steering techniques [10] and Inference-Time Intervention [2], we instead apply causal subtractive ablation. By mathematically subtracting the sycophantic directional component at layer 28, we successfully recover internal truth from the residual stream. Crucially, we utilize simultaneous block-patching (layers 20–30) to directly neutralize the "Hydra effect" [4], wherein early-layer interventions are overwritten by subsequent layers, achieving highly robust truth circuit localization superior to single-layer interventions.

4. REFERENCES

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